# DUNMAN SECONDARY SCHOOL

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| **CANDIDATE NAME** | | | |
| **CLASS** | | **INDEX NUMBER** | |

## PRELIMINARY EXAMINATION 2023
## SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

## ADDITIONAL MATHEMATICS — 4049/01

Paper 1

23 August 2023
2 hours 15 minutes

Candidates answer on the Question Paper.

**READ THESE INSTRUCTIONS FIRST**

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

| **Calculator Model** |
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Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an approved scientific calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 90.

| **Total Marks** |
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***Mathematical Formulae***

## 1. ALGEBRA

*Quadratic Equation*

For the equation $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

*Binomial expansion*

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where $n$ is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$.

## 2. TRIGONOMETRY

*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

*Formulae for $\Delta ABC$*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area of } \Delta = \frac{1}{2} ab \sin C$$

Answer **all** the questions.

**1** **(a)** Given that $\left[\left(\dfrac{20}{3}\right)^{-3} \times \sqrt{3^5}\right] \div \dfrac{3}{2} = 2^a 3^b 5^c$, find the value of each of $a$, $b$ and $c$. [3]

**(b)** It is given that $2^{2x+1} + 4^{x-1} = 2(3^{1-x})$.

Show that $12^x = 2\dfrac{2}{3}$. [3]

**2** **(a)** Solve $\log_2 x^2 = 3\log_x 2 + 1$. [4]

**(b)** Given that $\ln a - 2\ln b = \ln(a+b)$, express $a$ in terms of $b$. [3]

**3** **(a)** **Without using a calculator,** given that $\left(\sqrt{5}+1\right)x=\sqrt{3}$, express $x+\dfrac{1}{x}$ in the form of $a\sqrt{15}+b\sqrt{3}$, where $a$ and $b$ are rational numbers. [3]

**(b)** Given that $2\sqrt{2}-3=\dfrac{\sqrt{h-k\sqrt{2}}}{1+\sqrt{2}}$, find the values of $h$ and $k$. [3]

**4** It is given that $x-4$ is a factor of the polynomial $f(x) = 9x^3 + mx^2 - 23x + n$ where $m$ and $n$ are constants. $f(x)$ leaves a remainder of $-20$ when it is divided by $x+1$.

**(a)** Find the values of $m$ and $n$. [4]

**(b)** Using the values of $m$ and $n$ found in part **(a)**, solve the equation $\mathrm{f}(x) = 0$. [4]

**(c)** Hence use your answers to part **(b)** to solve the equation $72x^6 + 4mx^4 - 46x^2 + n = 0$. [2]

**5** **(a)** Express $\dfrac{11x^2-8x+3}{(x-1)(x^2+1)}$ in partial fractions. [5]

**(b)** Hence find $\displaystyle\int \frac{11x^2-8x+3}{(x-1)(x^2+1)}\,\mathrm{d}x$ [3]

**6** It is given that $f'(x) = \sin 3x - \cos \frac{x}{2}$. Given that $f\left(\frac{\pi}{3}\right) = 0$, show that

$$12f''(x) + 3f(x) = 35\cos 3x + 2\,.$$ [7]

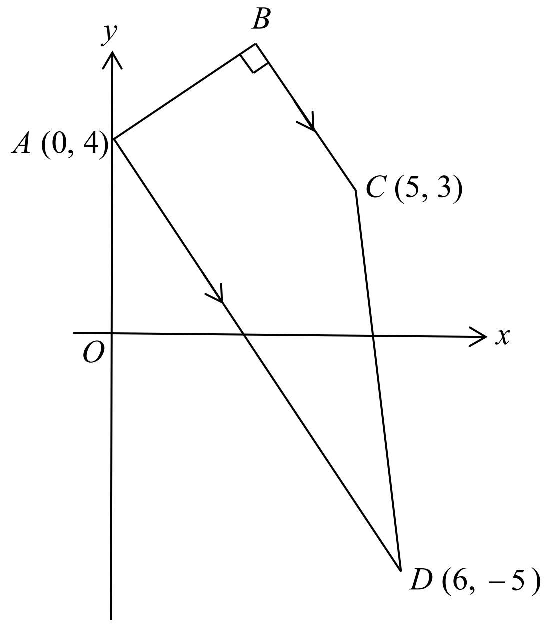
**7** **(a)** The expansion of $\left(1+y-y^2\right)^n$ in ascending powers of $y$ is $1+ay+20y^2+\ldots$.
Find the value of $n$ and of $a$. [4]

**(b)** In the expansion of $\left(2x^2 + \frac{a}{x}\right)^8$, where $a$ is a non-zero real number, the ratio of the coefficient of $x^{10}$ to that of the coefficient of $x^4$ is 5 : 2.

**(i)** Find the possible values of $a$. [5]

**(ii)** Explain whether the term independent of $x$ exists for the expansion of $\left(2x^2 + \frac{a}{x}\right)^8$. [2]

**8**




The diagram shows a trapezium $ABCD$ in which $AD$ is parallel to $BC$ and angle $ABC = 90°$.

The coordinates of $A$, $C$ and $D$ are (0, 4), (5, 3) and (6, – 5) respectively.

**(a)** Find the coordinates of $B$. [4]

**(b)** The point $E$ lies on the line $DA$ produced such that $\dfrac{\text{area of } \Delta BED}{\text{area of } \Delta BAD} = \dfrac{4}{3}$.

Show that the coordinates of point $E$ is $(-2,\ 7)$. [2]

**(c)** $F$ is a point such that $FECD$ is a rhombus.
Given that the coordinates of the midpoint of $FC$ is $(k+1,\ k)$, where $k$ is a constant, find the coordinates of $F$. [2]

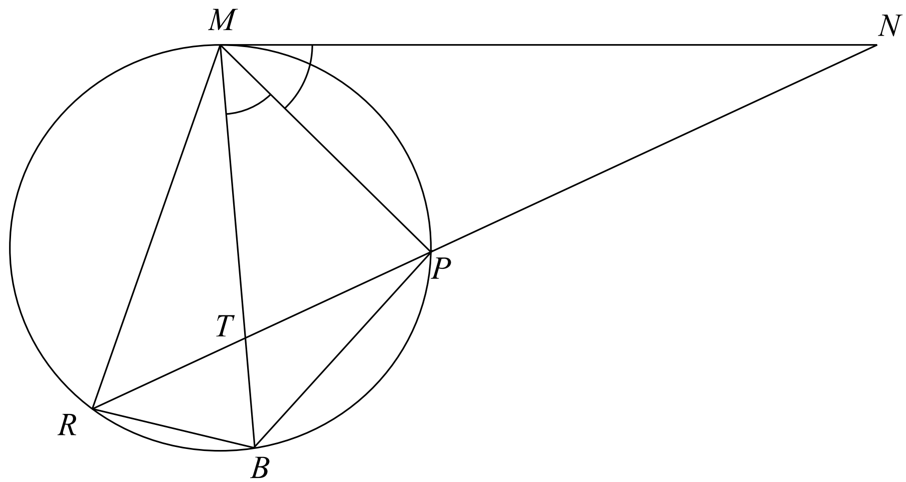
**9** **(a)** Find the set of values of the constant $k$ for which the curve $y = kx^2 + 4x + 2k$ lies entirely below the line $y = x + k$. [4]

**(b)** Show that there are no values of $m$ for which the curve $y = (m-5)x^2 + 2mx + (m+3)$ is always positive. [3]

**10** **(a)** Show that $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\ln x}{4x}\right)=\frac{1-\ln x}{4x^2}$. [3]

**(b)** Hence integrate $\frac{\ln x}{x^2}$ with respect to $x$. [3]

**11** The diagram shows a point $M$ on a circle and $MN$ is a tangent to the circle. Points $P$, $B$ and $R$ lie on the circle such that $MP$ bisects angle $NMB$ and $NPR$ is a straight line. The lines $NR$ and $MB$ intersect at $T$.




**(a)** Prove that $RP$ bisects angle $MRB$. [3]

**(b)** Prove that triangle *RTM* is similar to triangle *RBP*. [3]

**12** **(a)** Show that $\sin 3x = \sin x(4\cos^2 x - 1)$. [3]

**(b)** Hence solve the equation $3\sin 3x = 16\cos x \sin x$ for $0 \leq x \leq 2\pi$. [5]

**End of Paper**